

Lab 14 Mini Project RNA Seq

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Background

In today's mini project we analyze a high throughput biological data that require further interpretation. There are many freely available tools for pathway or over-representation analysis. As of Nov 2017 Bioconductor alone has over 80 packages categorized under gene set enrichment and over 120 packages categorized under pathways.

The data for our hands-on session comes from GEO entry, the authors report on differential analysis of lung fibroblasts in responsive to loss of the developmental transcription factor H0XA1.

Data Import

Read counts and metadata CSV files.

```
metadata <- read.csv("GSE37704_metadata.csv")
countData <- read.csv("GSE37704_featurecounts.csv", row.names = 1)
```

Sannity Check

We have a peek into our metadata

```
head(metadata)
```

	id	condition
1	SRR493366	control_sirna
2	SRR493367	control_sirna
3	SRR493368	control_sirna
4	SRR493369	hoxa1_kd
5	SRR493370	hoxa1_kd
6	SRR493371	hoxa1_kd

```
head(countData)
```

	length	SRR493366	SRR493367	SRR493368	SRR493369	SRR493370
ENSG00000186092	918	0	0	0	0	0
ENSG00000279928	718	0	0	0	0	0
ENSG00000279457	1982	23	28	29	29	28
ENSG00000278566	939	0	0	0	0	0
ENSG00000273547	939	0	0	0	0	0
ENSG00000187634	3214	124	123	205	207	212
		SRR493371				
ENSG00000186092		0				
ENSG00000279928		0				
ENSG00000279457		46				
ENSG00000278566		0				
ENSG00000273547		0				
ENSG00000187634		258				

Q. Complete the code below to remove the troublesome first column from countData

```
# Note we need to remove the odd first $length col
countData <- countData[,-1]
head(countData)
```

	SRR493366	SRR493367	SRR493368	SRR493369	SRR493370	SRR493371
ENSG00000186092	0	0	0	0	0	0
ENSG00000279928	0	0	0	0	0	0
ENSG00000279457	23	28	29	29	28	46
ENSG00000278566	0	0	0	0	0	0
ENSG00000273547	0	0	0	0	0	0
ENSG00000187634	124	123	205	207	212	258

```
colnames(countData)
```

```
[1] "SRR493366" "SRR493367" "SRR493368" "SRR493369" "SRR493370" "SRR493371"
```

```
metadata$id
```

```
[1] "SRR493366" "SRR493367" "SRR493368" "SRR493369" "SRR493370" "SRR493371"
```

```
all(colnames(countData)==metadata$id)
```

```
[1] TRUE
```

Q. Complete the code below to filter countData to exclude genes (i.e. rows) where we have 0 read count across all samples (i.e. columns).

We can sum across the rows for each gene and if the answer is zero then we have zero counts for that gene in all experiments...

```
# Filter count data where you have 0 read count across all samples.
countData = countData[rowSums(countData) > 0, ]
head(countData)
```

	SRR493366	SRR493367	SRR493368	SRR493369	SRR493370	SRR493371
ENSG00000279457	23	28	29	29	28	46
ENSG00000187634	124	123	205	207	212	258
ENSG00000188976	1637	1831	2383	1226	1326	1504
ENSG00000187961	120	153	180	236	255	357
ENSG00000187583	24	48	65	44	48	64
ENSG00000187642	4	9	16	14	16	16

```
dim(countData)
```

```
[1] 15975      6
```

Setup DESeq Object

```
library(DESeq2)
```

Loading required package: S4Vectors

Loading required package: stats4

Loading required package: BiocGenerics

Loading required package: generics

Attaching package: 'generics'

The following objects are masked from 'package:base':

```
as.difftime, as.factor, as.ordered, intersect, is.element, setdiff,  
setequal, union
```

Attaching package: 'BiocGenerics'

The following objects are masked from 'package:stats':

```
IQR, mad, sd, var, xtabs
```

The following objects are masked from 'package:base':

```
anyDuplicated, aperm, append, as.data.frame, basename, cbind,  
colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,  
get, grep, grepl, is.unsorted, lapply, Map, mapply, match, mget,  
order, paste, pmax, pmax.int, pmin, pmin.int, Position, rank,  
rbind, Reduce, rownames, sapply, saveRDS, table, tapply, unique,  
unsplit, which.max, which.min
```

Attaching package: 'S4Vectors'

The following object is masked from 'package:utils':

findMatches

The following objects are masked from 'package:base':

expand.grid, I, unname

Loading required package: IRanges

Loading required package: GenomicRanges

Loading required package: Seqinfo

Loading required package: SummarizedExperiment

Loading required package: MatrixGenerics

Loading required package: matrixStats

Attaching package: 'MatrixGenerics'

The following objects are masked from 'package:matrixStats':

colAlls, colAnyNAs, colAnys, colAvgsPerRowSet, colCollapse,
colCounts, colCummaxs, colCummins, colCumprods, colCumsums,
colDiffs, colIQRDiffs, colIQRs, colLogSumExps, colMadDiffs,
colMads, colMaxs, colMeans2, colMedians, colMins, colOrderStats,
colProds, colQuantiles, colRanges, colRanks, colSdDiffs, colSds,
colSums2, colTabulates, colVarDiffs, colVars, colWeightedMads,
colWeightedMeans, colWeightedMedians, colWeightedSds,
colWeightedVars, rowAlls, rowAnyNAs, rowAnys, rowAvgsPerColSet,
rowCollapse, rowCounts, rowCummaxs, rowCummins, rowCumprods,
rowCumsums, rowDiffs, rowIQRDiffs, rowIQRs, rowLogSumExps,
rowMadDiffs, rowMads, rowMaxs, rowMeans2, rowMedians, rowMins,

```
rowOrderStats, rowProds, rowQuantiles, rowRanges, rowRanks,
rowSdDiffs, rowSds, rowSums2, rowTabulates, rowVarDiffs, rowVars,
rowWeightedMads, rowWeightedMeans, rowWeightedMedians,
rowWeightedSds, rowWeightedVars
```

Loading required package: Biobase

Welcome to Bioconductor

```
Vignettes contain introductory material; view with
'browseVignettes()'. To cite Bioconductor, see
'citation("Biobase")', and for packages 'citation("pkgname")'.
```

Attaching package: 'Biobase'

The following object is masked from 'package:MatrixGenerics':

```
rowMedians
```

The following objects are masked from 'package:matrixStats':

```
anyMissing, rowMedians
```

```
dds <- DESeqDataSetFromMatrix(countData = countData,
                              colData = metadata,
                              design = ~condition)
dds
```

```
class: DESeqDataSet
dim: 15975 6
metadata(1): version
assays(1): counts
rownames(15975): ENSG00000279457 ENSG00000187634 ... ENSG00000276345
               ENSG00000271254
rowData names(0):
colnames(6): SRR493366 SRR493367 ... SRR493370 SRR493371
colData names(2): id condition
```

Run DESeq Analysis Pipeline

Now we can run the DESeq analysis pipeline using the `dds` object that has all the inputs we need.

```
dds <- DESeq(dds)
```

estimating size factors

estimating dispersions

gene-wise dispersion estimates

mean-dispersion relationship

final dispersion estimates

fitting model and testing

```
res <- results(dds)
head(res)
```

log2 fold change (MLE): condition hoxa1 kd vs control sirna

Wald test p-value: condition hoxa1 kd vs control sirna

DataFrame with 6 rows and 6 columns

	baseMean	log2FoldChange	lfcSE	stat	pvalue
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
ENSG00000279457	29.9136	0.1792571	0.3248215	0.551863	5.81042e-01
ENSG00000187634	183.2296	0.4264571	0.1402658	3.040350	2.36304e-03
ENSG00000188976	1651.1881	-0.6927205	0.0548465	-12.630156	1.43993e-36
ENSG00000187961	209.6379	0.7297556	0.1318599	5.534326	3.12428e-08
ENSG00000187583	47.2551	0.0405765	0.2718928	0.149237	8.81366e-01
ENSG00000187642	11.9798	0.5428105	0.5215598	1.040744	2.97994e-01
	padj				
	<numeric>				
ENSG00000279457	6.86555e-01				
ENSG00000187634	5.15718e-03				
ENSG00000188976	1.76553e-35				
ENSG00000187961	1.13413e-07				
ENSG00000187583	9.19031e-01				
ENSG00000187642	4.03379e-01				

Extract the Results

Q. Call the `summary()` function on your results to get a sense of how many genes are up or down-regulated at the default 0.1 p-value cutoff.

```
summary(res)
```

```
out of 15975 with nonzero total read count
adjusted p-value < 0.1
LFC > 0 (up)      : 4349, 27%
LFC < 0 (down)    : 4396, 28%
outliers [1]      : 0, 0%
low counts [2]    : 1237, 7.7%
(mean count < 0)
[1] see 'cooksCutoff' argument of ?results
[2] see 'independentFiltering' argument of ?results
```

Data Visualization

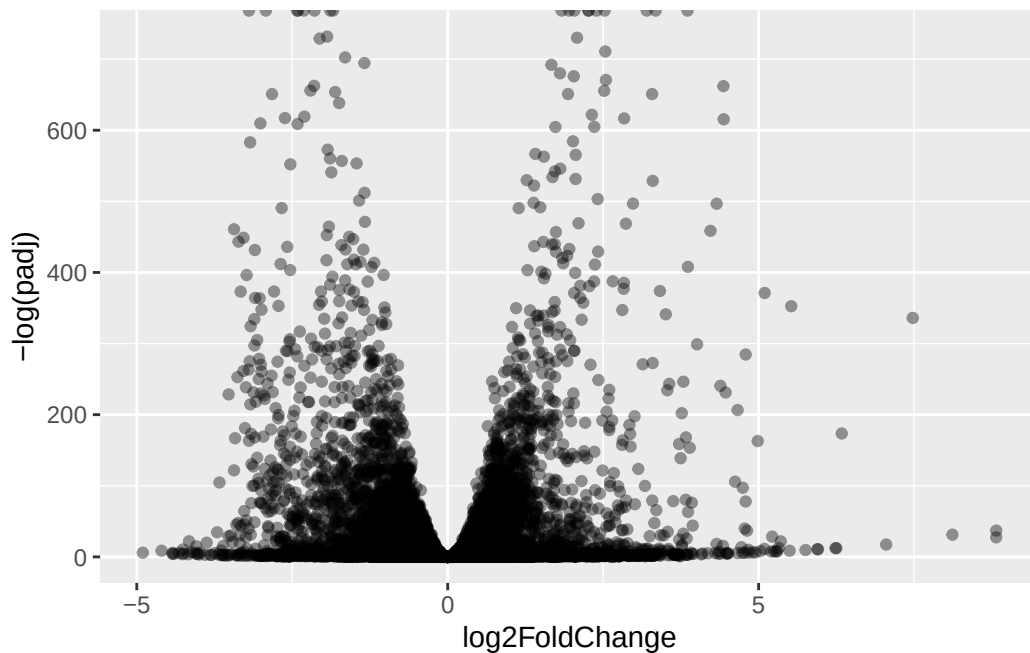
Volcano Plot

This is ubiquitous and common visualizaition for this type of data that puts the log2 fold change and the adjusted p-value together in one plot that people can get insight for what is going on in the whole dataset results.

```
library(ggplot2)
```

```
ggplot(res) +
  aes(log2FoldChange, -log(padj)) +
  geom_point(alpha = 0.4)
```

Warning: Removed 1237 rows containing missing values or values outside the scale range (``geom_point()``).



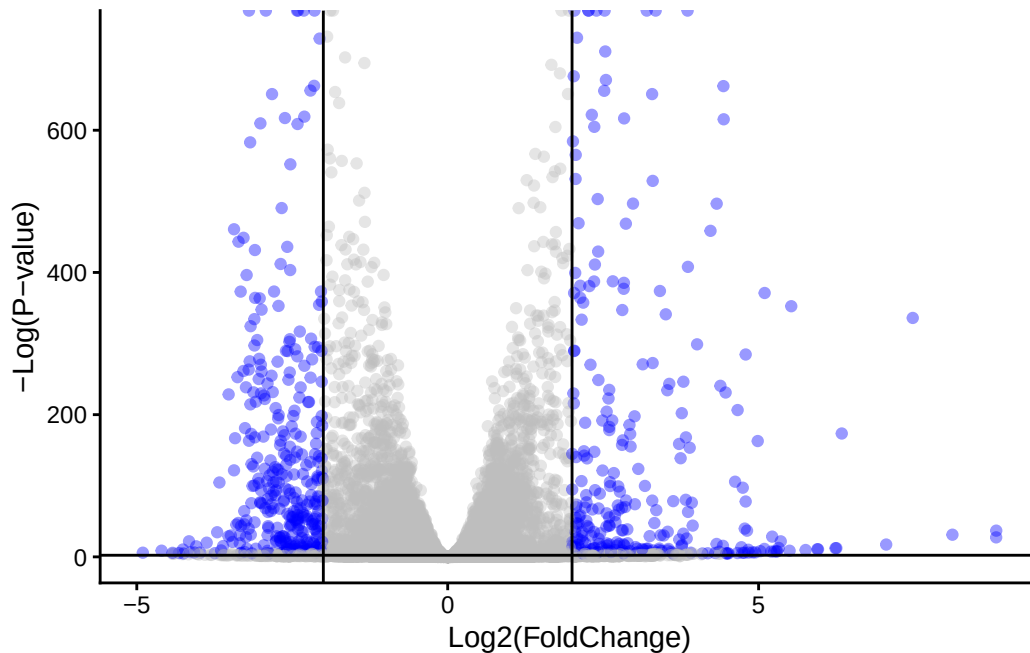
Q. Improve this plot by completing the below code, which adds color, axis labels and cutoff lines:

```
# Setup custom colors
mycols <- rep("gray", nrow(res))
mycols[abs(res$log2FoldChange) > 2] <- "blue"

mycols[res$padj > 0.01] <- "gray"

# Volcano plot with ggplot
ggplot(res) +
  aes(x = log2FoldChange, y = -log(padj)) +
  geom_point(color = mycols, alpha= 0.4) +
  geom_vline(xintercept = c(-2, 2),
             color = "black") +
  geom_hline(yintercept = -log(0.1),
             color = "black") +
  xlab("Log2(FoldChange)") +
  ylab("-Log(P-value)") +
  theme_classic()
```

Warning: Removed 1237 rows containing missing values or values outside the scale range (`geom\_point()`).



Add Annotation Data

Add gene symbol and entrez ids

Q. Use the `mapIDs()` function multiple times to add SYMBOL, ENTREZID and GENENAME annotation to our results by completing the code below.

```
library("AnnotationDbi")
library("org.Hs.eg.db")
```

```
columns(org.Hs.eg.db)
```

```
[1] "ACCNUM"      "ALIAS"       "ENSEMBL"     "ENSEMBLPROT" "ENSEMBLTRANS"
[6] "ENTREZID"    "ENZYME"      "EVIDENCE"    "EVIDENCEALL"  "GENENAME"
[11] "GENETYPE"    "GO"          "GOALL"       "IPI"          "MAP"
[16] "OMIM"        "ONTOLOGY"    "ONTOLOGYALL" "PATH"         "PFAM"
[21] "PMID"        "PROSITE"     "REFSEQ"      "SYMBOL"       "UCSCKG"
[26] "UNIPROT"
```

```
res$symbol = mapIds(org.Hs.eg.db,
                    keys=row.names(res),
                    keytype="ENSEMBL",
                    column="SYMBOL",
                    multiVals="first")
```

'select()' returned 1:many mapping between keys and columns

```
res$entrez = mapIds(org.Hs.eg.db,
                    keys=row.names(res),
                    keytype="ENSEMBL",
                    column="ENTREZID",
                    multiVals="first")
```

'select()' returned 1:many mapping between keys and columns

```
res$name = mapIds(org.Hs.eg.db,
                  keys=row.names(res),
                  keytype="ENSEMBL",
                  column="GENENAME",
                  multiVals="first")
```

'select()' returned 1:many mapping between keys and columns

```
head(res, 10)
```

log2 fold change (MLE): condition hoxa1 kd vs control sirna

Wald test p-value: condition hoxa1 kd vs control sirna

DataFrame with 10 rows and 9 columns

	baseMean	log2FoldChange	lfcSE	stat	pvalue
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
ENSG00000279457	29.913579	0.1792571	0.3248215	0.551863	5.81042e-01
ENSG00000187634	183.229650	0.4264571	0.1402658	3.040350	2.36304e-03
ENSG00000188976	1651.188076	-0.6927205	0.0548465	-12.630156	1.43993e-36
ENSG00000187961	209.637938	0.7297556	0.1318599	5.534326	3.12428e-08
ENSG00000187583	47.255123	0.0405765	0.2718928	0.149237	8.81366e-01
ENSG00000187642	11.979750	0.5428105	0.5215598	1.040744	2.97994e-01
ENSG00000188290	108.922128	2.0570638	0.1969053	10.446970	1.51281e-25
ENSG00000187608	350.716868	0.2573837	0.1027266	2.505522	1.22271e-02

ENSG00000188157	9128.439422	0.3899088	0.0467164	8.346302	7.04333e-17
ENSG00000237330	0.158192	0.7859552	4.0804729	0.192614	8.47261e-01
	padj	symbol	entrez		name
	<numeric>	<character>	<character>		<character>
ENSG00000279457	6.86555e-01	NA	NA		NA
ENSG00000187634	5.15718e-03	SAMD11	148398	sterile alpha motif	..
ENSG00000188976	1.76553e-35	NOC2L	26155	NOC2 like nucleolar	..
ENSG00000187961	1.13413e-07	KLHL17	339451	kelch like family me..	
ENSG00000187583	9.19031e-01	PLEKHN1	84069	pleckstrin homology	..
ENSG00000187642	4.03379e-01	PERM1	84808	PPARGC1 and ESRR	ind..
ENSG00000188290	1.30538e-24	HES4	57801	hes family bHLH tran..	
ENSG00000187608	2.37452e-02	ISG15	9636	ISG15 ubiquitin like..	
ENSG00000188157	4.21970e-16	AGRN	375790		agrin
ENSG00000237330	NA	RNF223	401934	ring finger protein	..

Q. Finally for this section let's reorder these results by adjusted p-value and save them to a CSV file in your current project directory.

```
res = res[order(res$pvalue),]
write.csv(res, file = "deseq_results.csv")
```

Pathway Analysis

KEGG Here we use the “gage” package for pathway analysis and once we have a list of enriched pathways, we're going to use the pathview package to draw pathway diagrams, shading the molecules in the pathway by their degree of up/down-regulation.

```
library(pathview)
```

```
#####
Pathview is an open source software package distributed under GNU General
Public License version 3 (GPLv3). Details of GPLv3 is available at
http://www.gnu.org/licenses/gpl-3.0.html. Particullary, users are required to
formally cite the original Pathview paper (not just mention it) in publications
or products. For details, do citation("pathview") within R.
```

The pathview downloads and uses KEGG data. Non-academic uses may require a KEGG license agreement (details at <http://www.kegg.jp/kegg/legal.html>).

```
#####
```

```
library(gage)
```

```
library(gageData)

data(kegg.sets.hs)
data(sigmet.idx.hs)

# Focus on signaling and metabolic pathways only
kegg.sets.hs = kegg.sets.hs[sigmet.idx.hs]

# Examine the first 3 pathways
head(kegg.sets.hs, 3)
```

```
$`hsa00232 Caffeine metabolism`
```

```
[1] "10" "1544" "1548" "1549" "1553" "7498" "9"
```

```
$`hsa00983 Drug metabolism - other enzymes`
```

```
[1] "10" "1066" "10720" "10941" "151531" "1548" "1549" "1551"
[9] "1553" "1576" "1577" "1806" "1807" "1890" "221223" "2990"
[17] "3251" "3614" "3615" "3704" "51733" "54490" "54575" "54576"
[25] "54577" "54578" "54579" "54600" "54657" "54658" "54659" "54963"
[33] "574537" "64816" "7083" "7084" "7172" "7363" "7364" "7365"
[41] "7366" "7367" "7371" "7372" "7378" "7498" "79799" "83549"
[49] "8824" "8833" "9" "978"
```

```
$`hsa00230 Purine metabolism`
```

```
[1] "100" "10201" "10606" "10621" "10622" "10623" "107" "10714"
[9] "108" "10846" "109" "111" "11128" "11164" "112" "113"
[17] "114" "115" "122481" "122622" "124583" "132" "158" "159"
[25] "1633" "171568" "1716" "196883" "203" "204" "205" "221823"
[33] "2272" "22978" "23649" "246721" "25885" "2618" "26289" "270"
[41] "271" "27115" "272" "2766" "2977" "2982" "2983" "2984"
[49] "2986" "2987" "29922" "3000" "30833" "30834" "318" "3251"
[57] "353" "3614" "3615" "3704" "377841" "471" "4830" "4831"
[65] "4832" "4833" "4860" "4881" "4882" "4907" "50484" "50940"
[73] "51082" "51251" "51292" "5136" "5137" "5138" "5139" "5140"
[81] "5141" "5142" "5143" "5144" "5145" "5146" "5147" "5148"
[89] "5149" "5150" "5151" "5152" "5153" "5158" "5167" "5169"
[97] "51728" "5198" "5236" "5313" "5315" "53343" "54107" "5422"
```

```
[105] "5424" "5425" "5426" "5427" "5430" "5431" "5432" "5433"
[113] "5434" "5435" "5436" "5437" "5438" "5439" "5440" "5441"
[121] "5471" "548644" "55276" "5557" "5558" "55703" "55811" "55821"
[129] "5631" "5634" "56655" "56953" "56985" "57804" "58497" "6240"
[137] "6241" "64425" "646625" "654364" "661" "7498" "8382" "84172"
[145] "84265" "84284" "84618" "8622" "8654" "87178" "8833" "9060"
[153] "9061" "93034" "953" "9533" "954" "955" "956" "957"
[161] "9583" "9615"
```

```
foldchanges = res$log2FoldChange
names(foldchanges) = res$entrez
head(foldchanges)
```

```
      1266      54855      1465      2034      2150      6659
-2.422719  3.201955 -2.313738 -1.888019  3.344508  2.392288
```

```
# Get the results
keggres = gage(foldchanges, gsets=kegg.sets.hs)
```

Now lets look at the object returned from gage().

```
attributes(keggres)
```

```
$names
[1] "greater" "less" "stats"
```

```
# Look at the first few down (less) pathways
head(keggres$less)
```

	p.geomean	stat.mean	p.val
hsa04110 Cell cycle	8.995727e-06	-4.378644	8.995727e-06
hsa03030 DNA replication	9.424076e-05	-3.951803	9.424076e-05
hsa03013 RNA transport	1.375901e-03	-3.028500	1.375901e-03
hsa03440 Homologous recombination	3.066756e-03	-2.852899	3.066756e-03
hsa04114 Oocyte meiosis	3.784520e-03	-2.698128	3.784520e-03
hsa00010 Glycolysis / Gluconeogenesis	8.961413e-03	-2.405398	8.961413e-03

	q.val	set.size	exp1
hsa04110 Cell cycle	0.001448312	121	8.995727e-06
hsa03030 DNA replication	0.007586381	36	9.424076e-05
hsa03013 RNA transport	0.073840037	144	1.375901e-03

hsa03440	Homologous recombination	0.121861535	28	3.066756e-03
hsa04114	Oocyte meiosis	0.121861535	102	3.784520e-03
hsa00010	Glycolysis / Gluconeogenesis	0.212222694	53	8.961413e-03

Now, let's try out the `pathview()` function from the **pathview** package to make a pathway plot with our RNA-Seq expression results shown in color. This downloads the pathway figure data from KEGG and adds our results to it. Here is the default low resolution raster PNG output from the `pathview()` call above

```
pathview(gene.data=foldchanges, pathway.id="hsa04110")
```

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/madinakho/Desktop/UCSD/Classes/Spring 26/BIMM 143/Class R/1

Info: Writing image file hsa04110.pathview.png


```
## Focus on top 5 upregulated pathways here for demo purposes only
keggrespathways <- rownames(keggres$greater)[1:5]

# Extract the 8 character long IDs part of each string
keggresids = substr(keggrespathways, start=1, stop=8)
keggresids
```

```
[1] "hsa04640" "hsa04630" "hsa00140" "hsa04142" "hsa04330"
```

```
pathview(gene.data=foldchanges, pathway.id=keggresids, species="hsa")
```

```
'select()' returned 1:1 mapping between keys and columns
```

```
Info: Working in directory /Users/madinakho/Desktop/UCSD/Classes/Spring 26/BIMM 143/Class R/1
```

```
Info: Writing image file hsa04640.pathview.png
```

```
'select()' returned 1:1 mapping between keys and columns
```

```
Info: Working in directory /Users/madinakho/Desktop/UCSD/Classes/Spring 26/BIMM 143/Class R/1
```

```
Info: Writing image file hsa04630.pathview.png
```

```
'select()' returned 1:1 mapping between keys and columns
```

```
Info: Working in directory /Users/madinakho/Desktop/UCSD/Classes/Spring 26/BIMM 143/Class R/1
```

```
Info: Writing image file hsa00140.pathview.png
```

```
'select()' returned 1:1 mapping between keys and columns
```

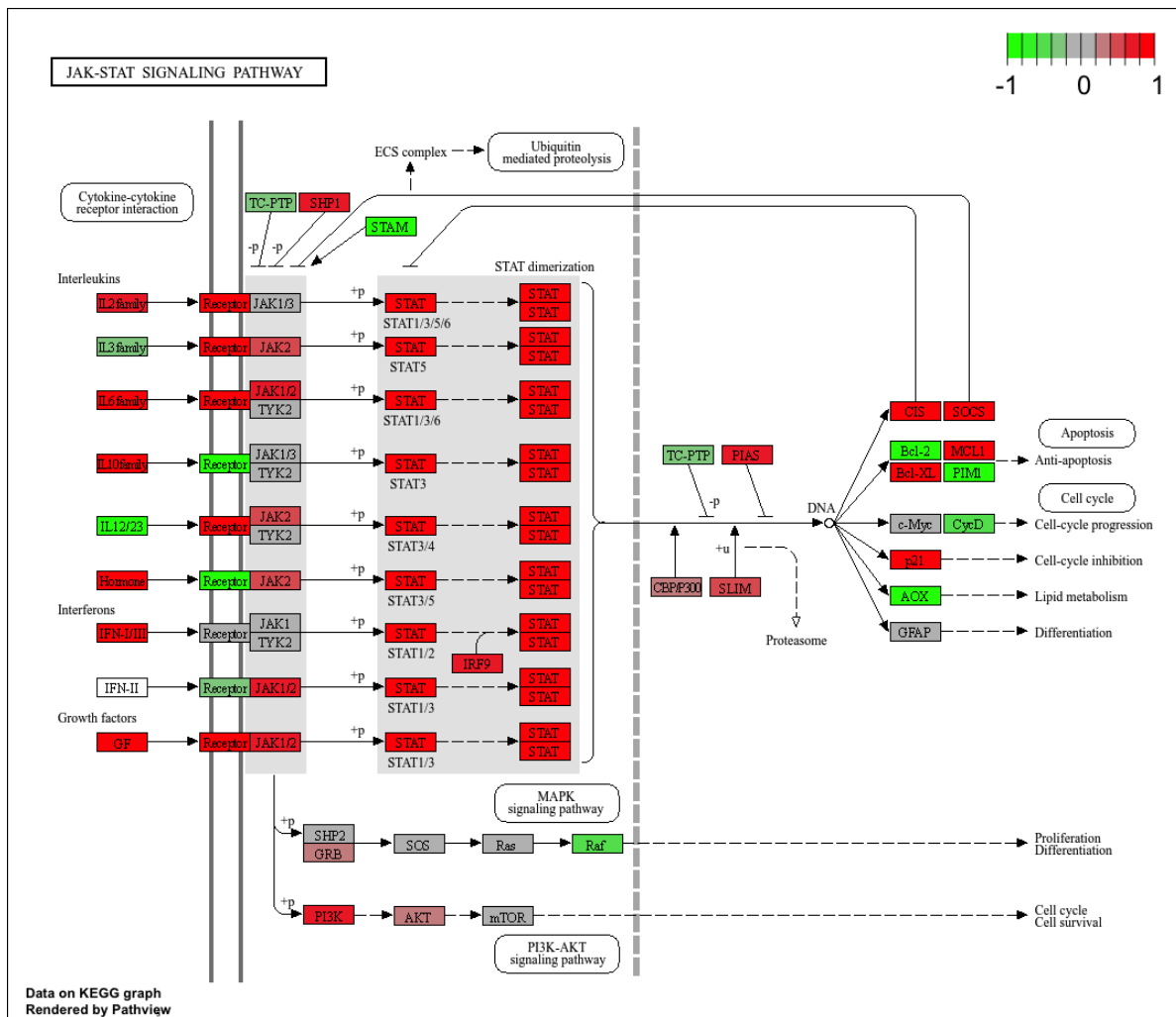
```
Info: Working in directory /Users/madinakho/Desktop/UCSD/Classes/Spring 26/BIMM 143/Class R/1
```

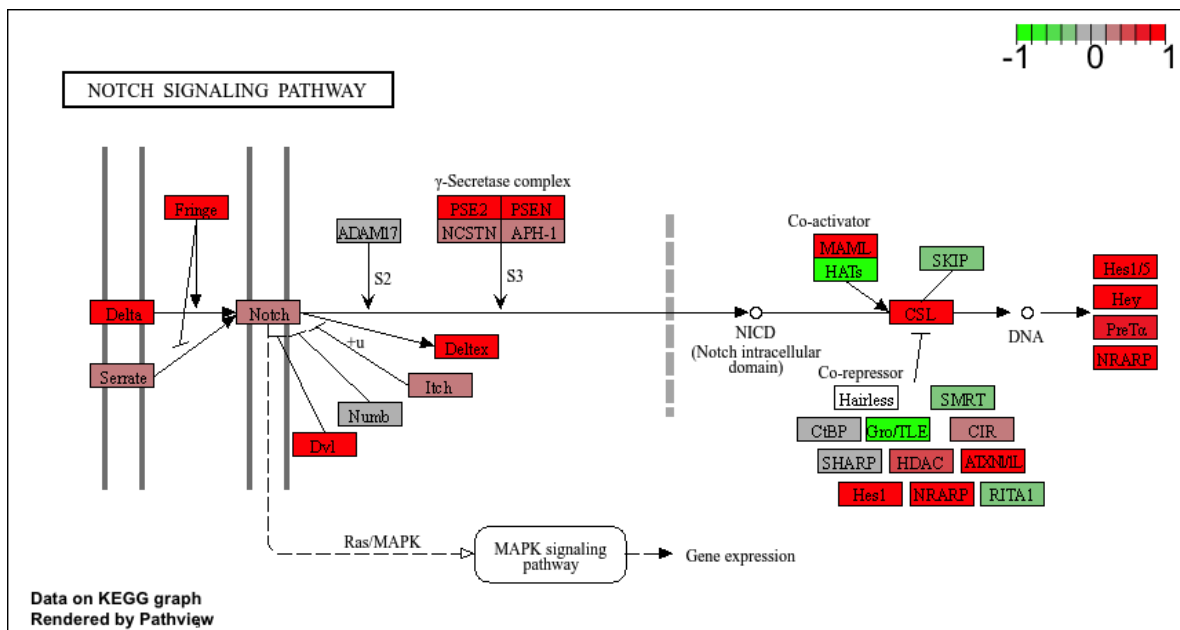
```
Info: Writing image file hsa04142.pathview.png
```

```
'select()' returned 1:1 mapping between keys and columns
```

Info: Working in directory /Users/madinakho/Desktop/UCSD/Classes/Spring 26/BIMM 143/Class R/1

Info: Writing image file hsa04330.pathview.png





Q. Can you do the same procedure as above to plot the pathview figures for the top 5 down-regulated pathways?

```
keggrespathways <- rownames(keggres$less)[1:5]
```

```
# Extract the 8 character long IDs part of each string
keggresids = substr(keggrespathways, start=1, stop=8)
keggresids
```

```
[1] "hsa04110" "hsa03030" "hsa03013" "hsa03440" "hsa04114"
```

```
pathview(gene.data=foldchanges, pathway.id=keggresids, species="hsa")
```

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/madinakho/Desktop/UCSD/Classes/Spring 26/BIMM 143/Class R/1

Info: Writing image file hsa04110.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/madinakho/Desktop/UCSD/Classes/Spring 26/BIMM 143/Class R/1

Info: Writing image file hsa03030.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/madinakho/Desktop/UCSD/Classes/Spring 26/BIMM 143/Class R/1

Info: Writing image file hsa03013.pathview.png

'select()' returned 1:1 mapping between keys and columns

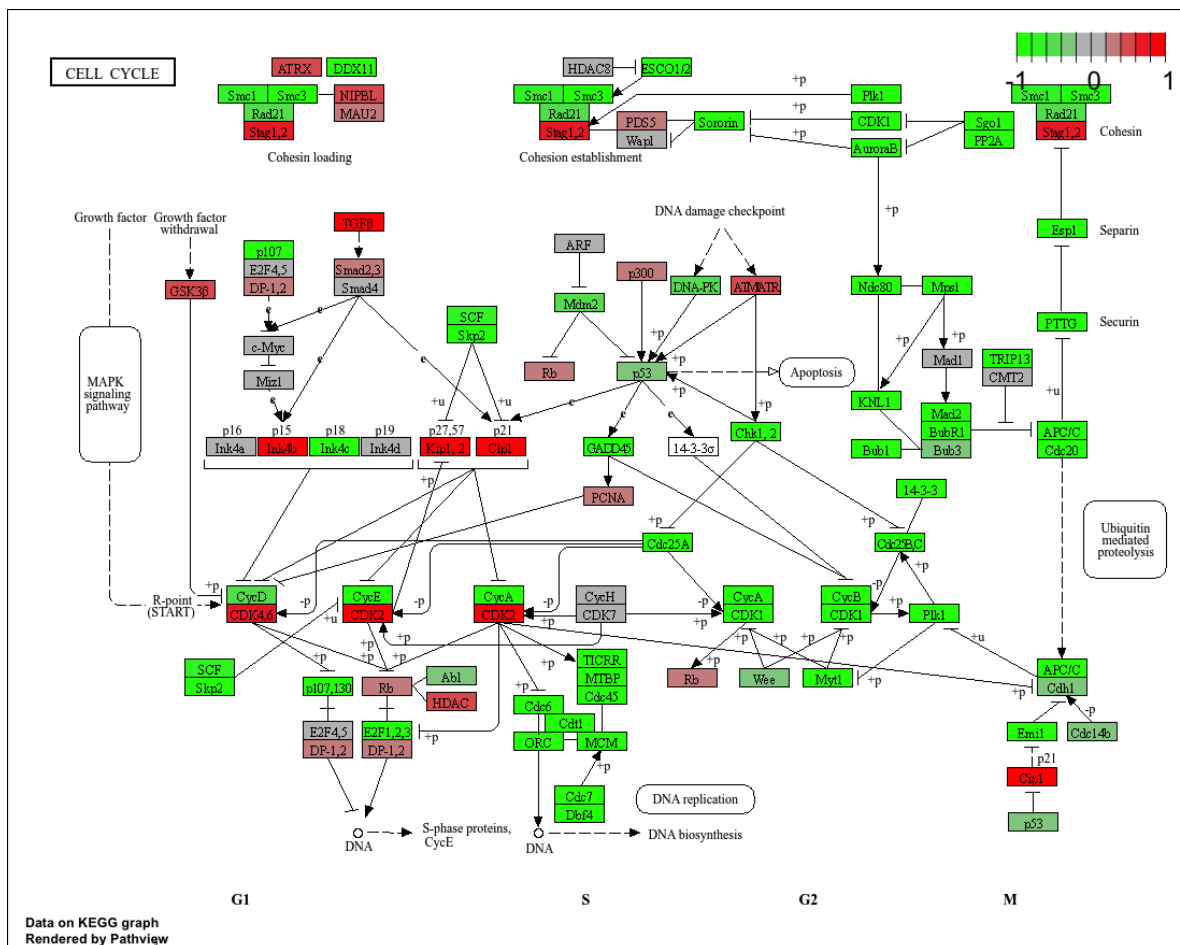
Info: Working in directory /Users/madinakho/Desktop/UCSD/Classes/Spring 26/BIMM 143/Class R/1

Info: Writing image file hsa03440.pathview.png

'select()' returned 1:1 mapping between keys and columns

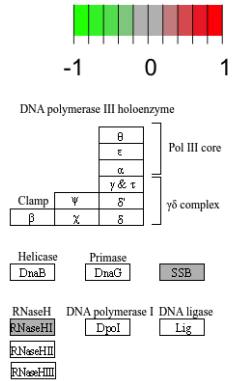
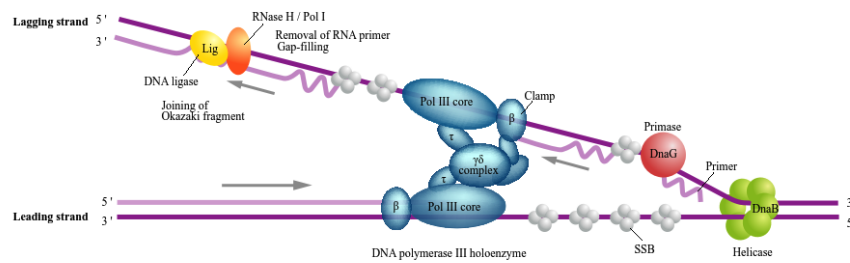
Info: Working in directory /Users/madinakho/Desktop/UCSD/Classes/Spring 26/BIMM 143/Class R/1

Info: Writing image file hsa04114.pathview.png

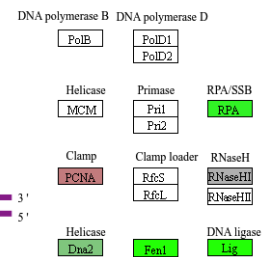
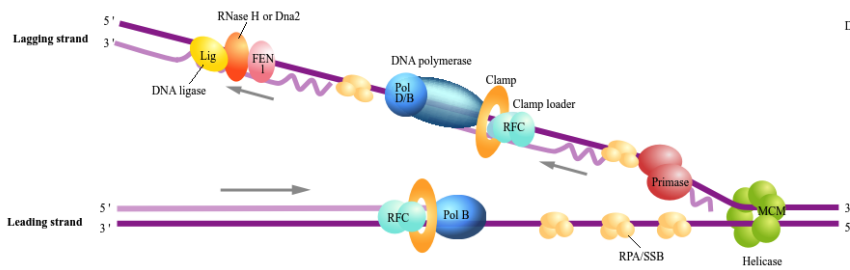


DNA REPLICATION

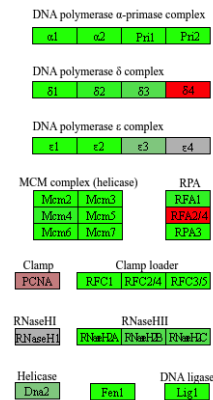
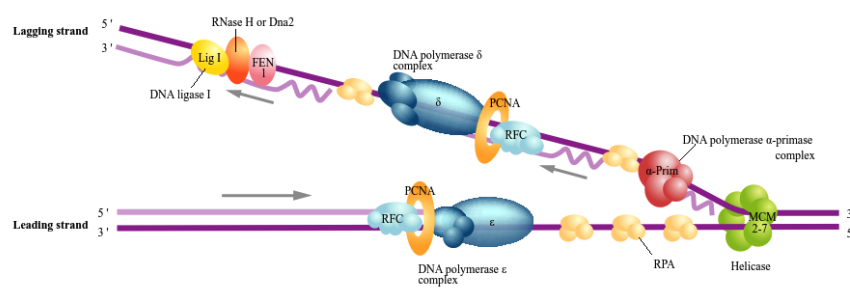
Replication complex (Bacteria)



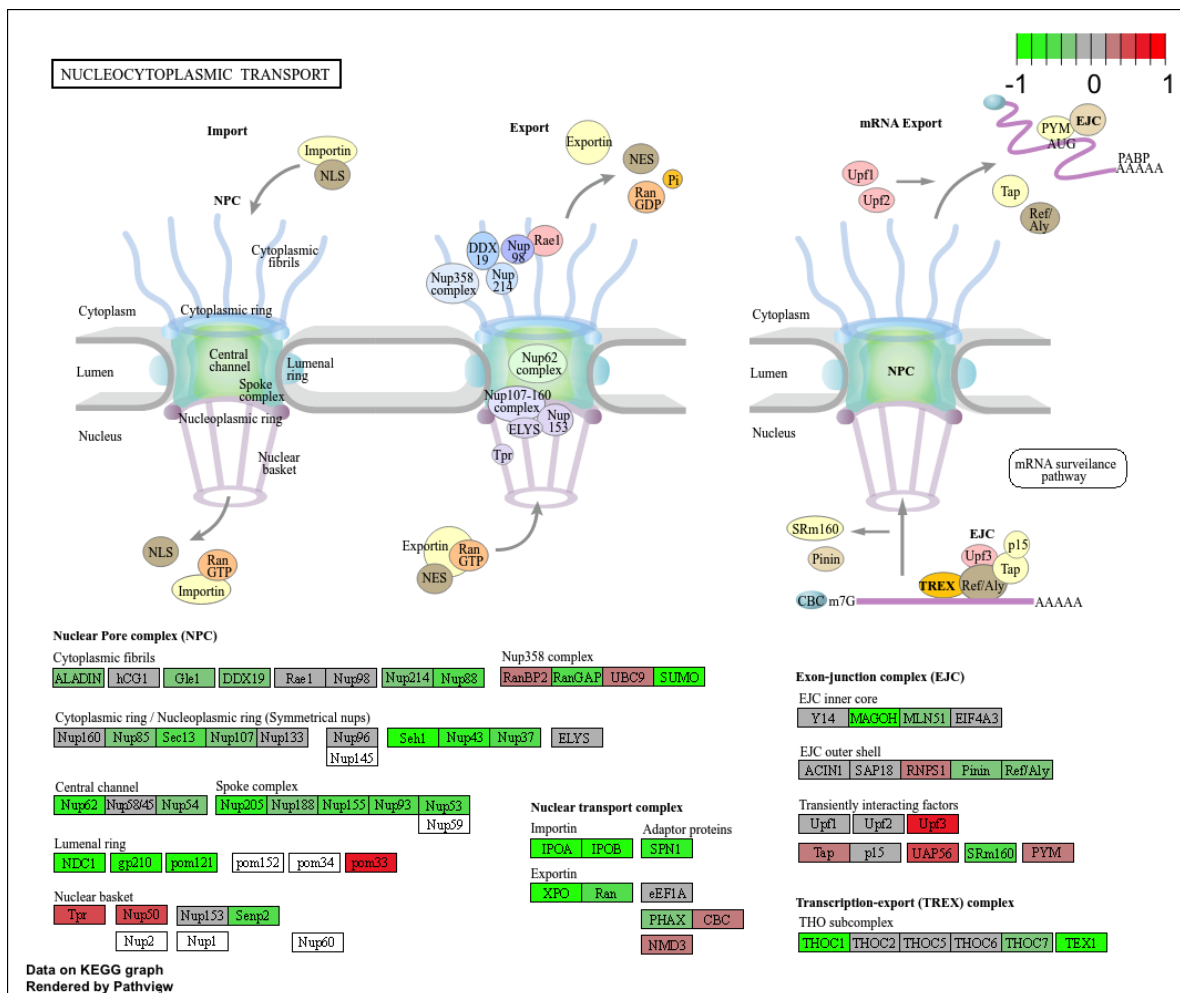
Replication complex (Archaea)

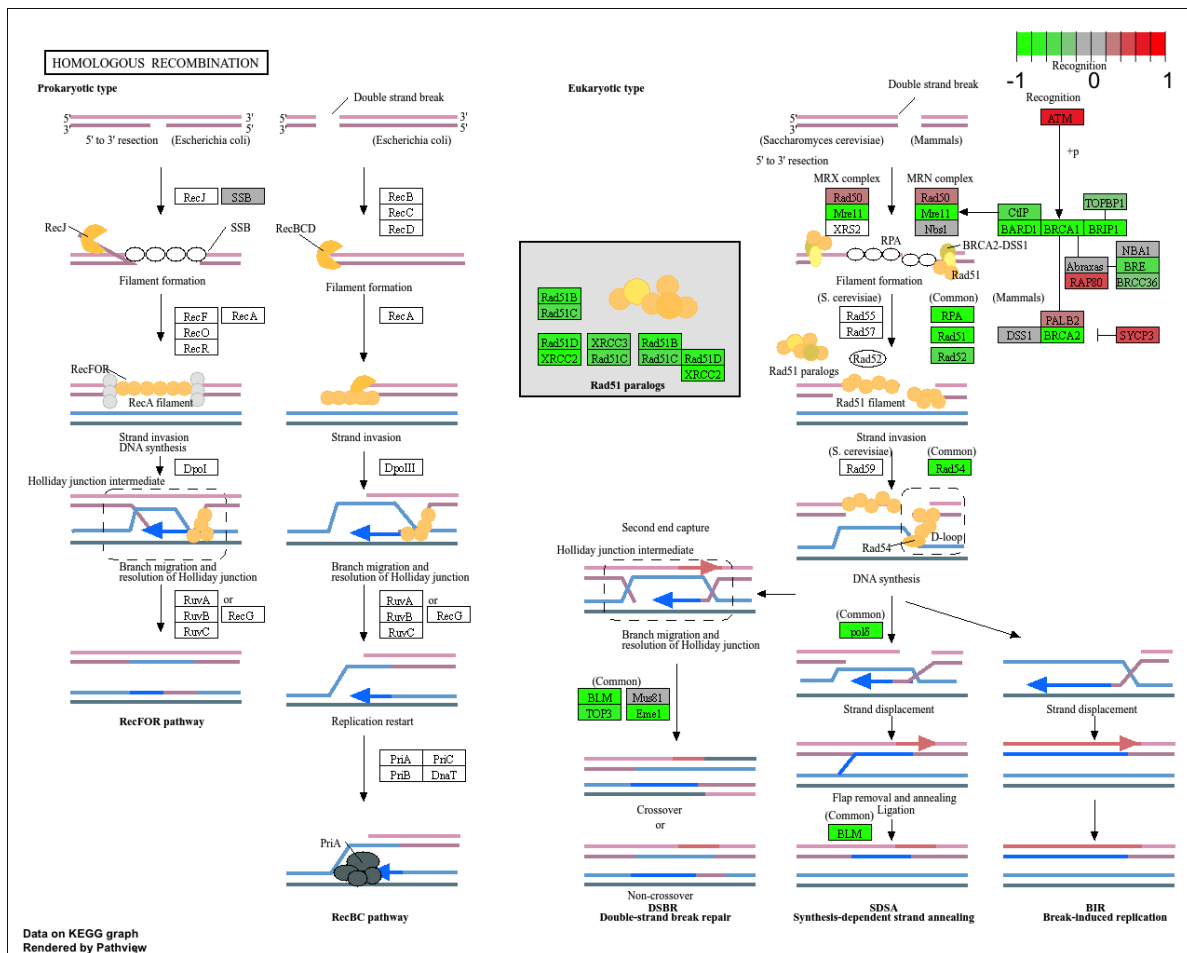


Replication complex (Eukaryotes)



Data on KEGG graph
Rendered by Pathview





	p.geomean	stat.mean	p.val
G0:0007156 homophilic cell adhesion	8.519724e-05	3.824205	8.519724e-05
G0:0002009 morphogenesis of an epithelium	1.396681e-04	3.653886	1.396681e-04
G0:0048729 tissue morphogenesis	1.432451e-04	3.643242	1.432451e-04
G0:0007610 behavior	1.925222e-04	3.565432	1.925222e-04
G0:0060562 epithelial tube morphogenesis	5.932837e-04	3.261376	5.932837e-04
G0:0035295 tube development	5.953254e-04	3.253665	5.953254e-04

	q.val	set.size	exp1
G0:0007156 homophilic cell adhesion	0.1951953	113	8.519724e-05
G0:0002009 morphogenesis of an epithelium	0.1951953	339	1.396681e-04
G0:0048729 tissue morphogenesis	0.1951953	424	1.432451e-04
G0:0007610 behavior	0.1967577	426	1.925222e-04
G0:0060562 epithelial tube morphogenesis	0.3565320	257	5.932837e-04
G0:0035295 tube development	0.3565320	391	5.953254e-04

\$less

	p.geomean	stat.mean	p.val
G0:0048285 organelle fission	1.536227e-15	-8.063910	1.536227e-15
G0:0000280 nuclear division	4.286961e-15	-7.939217	4.286961e-15
G0:0007067 mitosis	4.286961e-15	-7.939217	4.286961e-15
G0:0000087 M phase of mitotic cell cycle	1.169934e-14	-7.797496	1.169934e-14
G0:0007059 chromosome segregation	2.028624e-11	-6.878340	2.028624e-11
G0:0000236 mitotic prometaphase	1.729553e-10	-6.695966	1.729553e-10

	q.val	set.size	exp1
G0:0048285 organelle fission	5.841698e-12	376	1.536227e-15
G0:0000280 nuclear division	5.841698e-12	352	4.286961e-15
G0:0007067 mitosis	5.841698e-12	352	4.286961e-15
G0:0000087 M phase of mitotic cell cycle	1.195672e-11	362	1.169934e-14
G0:0007059 chromosome segregation	1.658603e-08	142	2.028624e-11
G0:0000236 mitotic prometaphase	1.178402e-07	84	1.729553e-10

\$stats

	stat.mean	exp1
G0:0007156 homophilic cell adhesion	3.824205	3.824205
G0:0002009 morphogenesis of an epithelium	3.653886	3.653886
G0:0048729 tissue morphogenesis	3.643242	3.643242
G0:0007610 behavior	3.565432	3.565432
G0:0060562 epithelial tube morphogenesis	3.261376	3.261376
G0:0035295 tube development	3.253665	3.253665

REACTOME

Let's now conduct over-representation enrichment analysis and pathway-topology analysis with Reactome using the previous list of significant genes generated from our differential expression results above.

```
## First, using R, output the list of significant genes at the 0.05 level as a plain text file

sig_genes <- res[res$padj <= 0.05 & !is.na(res$padj), "symbol"]
print(paste("Total number of significant genes:", length(sig_genes)))
```

```
[1] "Total number of significant genes: 8147"
```

```
write.table(sig_genes, file="significant_genes.txt", row.names=FALSE, col.names=FALSE, quote=
```

```
reactome <- read.csv("Reactome_results.csv")
head(reactome, 10)
```

	Pathway.identifier		Pathway.name
1	R-HSA-1640170		Cell Cycle
2	R-HSA-69278		Cell Cycle, Mitotic
3	R-HSA-69620		Cell Cycle Checkpoints
4	R-HSA-68877		Mitotic Prometaphase
5	R-HSA-69618		Mitotic Spindle Checkpoint
6	R-HSA-141424		Amplification of signal from the kinetochores
7	R-HSA-141444	Amplification of signal from unattached kinetochores via a MAD2 inhibitory signal	
8	R-HSA-927802		Nonsense-Mediated Decay (NMD)
9	R-HSA-975957	Nonsense Mediated Decay (NMD) enhanced by the Exon Junction Complex (EJC)	
10	R-HSA-453279		Mitotic G1 phase and G1/S transition
	X.Entities.found	X.Entities.total	Entities.ratio Entities.pValue

1	493	729	0.045030576	2.626230e-05
2	405	587	0.036259188	2.779371e-05
3	200	278	0.017172154	4.101809e-04
4	156	212	0.013095312	6.787224e-04
5	89	111	0.006856508	8.719206e-04
6	77	94	0.005806412	1.061383e-03
7	77	94	0.005806412	1.061383e-03
8	95	124	0.007659522	2.268941e-03
9	95	124	0.007659522	2.268941e-03
10	121	164	0.010130335	2.279467e-03

	Entities.FDR	X.Reactions.found	X.Reactions.total	Reactions.ratio
1	0.03835532	472	474	0.0296138948
2	0.03835532	367	367	0.0229289017
3	0.37736643	63	63	0.0039360240
4	0.41818475	25	25	0.0015619143
5	0.41818475	7	7	0.0004373360
6	0.41818475	4	4	0.0002499063
7	0.41818475	4	4	0.0002499063
8	0.60942115	6	6	0.0003748594
9	0.60942115	5	5	0.0003123829
10	0.60942115	101	101	0.0063101337

	Species.identifier	Species.name
1	9606	Homo sapiens
2	9606	Homo sapiens
3	9606	Homo sapiens
4	9606	Homo sapiens
5	9606	Homo sapiens
6	9606	Homo sapiens
7	9606	Homo sapiens
8	9606	Homo sapiens
9	9606	Homo sapiens
10	9606	Homo sapiens

1 RB1;MDC1;NUP107;POP1;SMC3;SMC4;BACH1;SMC2;PSMD8;PSMD6;SCP2;MYC;PSMD2;AKT2;PSMD3;AKT3;PSMD3A;RAB8A;NDEL1;H3-3B;SEC13;TUBB;KIF23;TUBG2;TUBG1;H2AJ;RAD51C;KIF2A;KIF2C;MAPRE1;SFI1;SDCCAGL1

2 3A;RAB8A;NDEL1;LIN52;H3-3B;NDC1;SEC13;NUDC;TUBB;KIF23;TUBG2;DYNLL1;TUBG1;DYNLL2;SIRT2;MZT2B1

3

4

5

6

7

8

9
10

Mapped.entities

1	NA
2	NA
3	NA
4	NA
5	NA
6	NA
7	NA
8	NA
9	NA
10	NA

1 R-HSA-156673;R-HSA-156678;R-HSA-156682;R-HSA-912389;R-HSA-174088;R-HSA-9929884;R-HSA-174097;R-HSA-912408;R-HSA-156699;R-HSA-174104;R-HSA-174105;R-HSA-174110;R-HSA-9928887;R-HSA-912429;R-HSA-75809;R-HSA-3000335;R-HSA-174119;R-HSA-9929904;R-HSA-174122;R-HSA-174120;R-HSA-3000327;R-HSA-174121;R-HSA-176175;R-HSA-75820;R-HSA-75823;R-HSA-174124;R-HSA-75822;R-HSA-75825;R-HSA-156723;R-HSA-75824;R-HSA-2470935;R-HSA-69685;R-HSA-174132;R-HSA-174139;R-HSA-9928875;R-HSA-170044;R-HSA-3000339;R-HSA-174144;R-HSA-170055;R-HSA-606287;R-HSA-912458;R-HSA-170057;R-HSA-9624798;R-HSA-9668831;R-HSA-174159;R-HSA-174157;R-HSA-912450;R-HSA-170070;R-HSA-174164;R-HSA-9929935;R-HSA-606289;R-HSA-174171;R-HSA-170072;R-HSA-8942803;R-HSA-912470;R-HSA-174174;R-HSA-170076;R-HSA-912467;R-HSA-141409;R-HSA-170087;R-HSA-170088;R-HSA-141422;R-HSA-141423;R-HSA-187506;R-HSA-174195;R-HSA-9624800;R-HSA-912505;R-HSA-141431;R-HSA-141429;R-HSA-174202;R-HSA-176250;R-HSA-174203;R-HSA-912503;R-HSA-8942836;R-HSA-141439;R-HSA-912496;R-HSA-141437;R-HSA-606326;R-HSA-187520;R-HSA-174209;R-HSA-606349;R-HSA-170116;R-HSA-8942607;R-HSA-8962050;R-HSA-176264;R-HSA-170120;R-HSA-170126;R-HSA-2995389;R-HSA-170131;R-HSA-174227;R-HSA-2995388;R-HSA-174224;R-HSA-6804724;R-HSA-174235;R-HSA-8848414;R-HSA-2473151;R-HSA-187545;R-HSA-174238;R-HSA-2995376;R-HSA-187552;R-HSA-170149;R-HSA-176298;R-HSA-174251;R-HSA-170153;R-HSA-2422927;R-HSA-170158;R-HSA-3000449;R-HSA-174255;R-HSA-170159;R-HSA-170156;R-HSA-170161;R-HSA-8848436;R-HSA-187574;R-HSA-187575;R-HSA-8932400;R-HSA-9851946;R-HSA-176318;R-HSA-163010;R-HSA-174273;R-HSA-181450;R-HSA-179410;R-HSA-157906;R-HSA-9851972;R-HSA-179417;R-HSA-179421;R-HSA-2468039;R-HSA-2473152;R-HSA-4088024;R-HSA-264418;R-HSA-2468041;R-HSA-2468040;R-HSA-349426;R-HSA-264435;R-HSA-264444;R-HSA-5693609;R-HSA-349444;R-HSA-69891;R-HSA-2172194;R-HSA-75010;R-HSA-8964492;R-HSA-75016;R-HSA-8964482;R-HSA-349455;R-HSA-264458;R-HSA-2984226;R-HSA-163090;R-HSA-75028;R-HSA-163099;R-HSA-163096;R-HSA-8964498;R-HSA-8964525;R-HSA-8964513;R-HSA-9853372;R-HSA-2168079;R-HSA-9853369;R-HSA-4419948;R-HSA-68913;R-HSA-2984220;R-HSA-2466068;R-HSA-163120;R-HSA-68914;R-HSA-68917;R-HSA-68916;R-HSA-2172183;R-HSA-2301205;R-

HSA-68919;R-HSA-68918;R-HSA-9634219;R-HSA-8964531;R-HSA-6803801;R-HSA-9659820;R-HSA-182594;R-HSA-1362261;R-HSA-9670101;R-HSA-1362270;R-HSA-9023941;R-HSA-68940;R-HSA-8964550;R-HSA-9023943;R-HSA-68944;R-HSA-68947;R-HSA-68946;R-HSA-68948;R-HSA-68950;R-HSA-1363276;R-HSA-174427;R-HSA-8964561;R-HSA-68954;R-HSA-174425;R-HSA-4088162;R-HSA-1363274;R-HSA-9853385;R-HSA-8964567;R-HSA-6803719;R-HSA-68960;R-HSA-6804741;R-HSA-8964588;R-HSA-174438;R-HSA-2314566;R-HSA-174439;R-HSA-4088152;R-HSA-1363314;R-HSA-8978926;R-HSA-9686521;R-HSA-2314569;R-HSA-174441;R-HSA-174446;R-HSA-8964580;R-HSA-174447;R-HSA-174444;R-HSA-174445;R-HSA-9686524;R-HSA-2984258;R-HSA-9670114;R-HSA-174451;R-HSA-174448;R-HSA-380278;R-HSA-4088141;R-HSA-1363303;R-HSA-380272;R-HSA-174452;R-HSA-4088134;R-HSA-174456;R-HSA-9671145;R-HSA-1363311;R-HSA-1226094;R-HSA-4088130;R-HSA-1226095;R-HSA-380283;R-HSA-1363306;R-HSA-380294;R-HSA-3002798;R-HSA-1638803;R-HSA-380303;R-HSA-69005;R-HSA-69006;R-HSA-380311;R-HSA-1363328;R-HSA-3002811;R-HSA-69015;R-HSA-1363331;R-HSA-380316;R-HSA-69016;R-HSA-2468287;R-HSA-69019;R-HSA-9624845;R-HSA-9624893;R-HSA-6793685;R-HSA-9668902;R-HSA-187828;R-HSA-9668904;R-HSA-1638821;R-HSA-6804955;R-HSA-69053;R-HSA-9624876;R-HSA-2288097;R-HSA-69063;R-HSA-4088309;R-HSA-4088306;R-HSA-4088307;R-HSA-69068;R-HSA-4088305;R-HSA-8941895;R-HSA-8941915;R-HSA-69074;R-HSA-4088298;R-HSA-4088299;R-HSA-2574845;R-HSA-2574840;R-HSA-2468293;R-HSA-5633460;R-HSA-9667952;R-HSA-6804879;R-HSA-9634169;R-HSA-9029987;R-HSA-2214351;R-HSA-9667965;R-HSA-8856951;R-HSA-8964475;R-HSA-8856945;R-HSA-169461;R-HSA-2562526;R-HSA-8964465;R-HSA-69116;R-HSA-169468;R-HSA-8964471;R-HSA-375302;R-HSA-2562594;R-HSA-69127;R-HSA-8961678;R-HSA-5683792;R-HSA-8961665;R-HSA-187916;R-HSA-8961671;R-HSA-8961688;R-HSA-9943687;R-HSA-9647746;R-HSA-2545203;R-HSA-69140;R-HSA-69142;R-HSA-69144;R-HSA-187934;R-HSA-69152;R-HSA-380455;R-HSA-187937;R-HSA-9018017;R-HSA-8961699;R-HSA-187948;R-HSA-187949;R-HSA-9943736;R-HSA-69173;R-HSA-187959;R-HSA-176702;R-HSA-176700;R-HSA-2245218;R-HSA-5683735;R-HSA-2545253;R-HSA-69191;R-HSA-4088441;R-HSA-4088439;R-HSA-69195;R-HSA-69199;R-HSA-380508;R-HSA-2430533;R-HSA-2430535;R-HSA-5683774;R-HSA-69227;R-HSA-2484822;R-HSA-9668335;R-HSA-2430552;R-HSA-9754129;R-HSA-9754130;R-HSA-9615901;R-HSA-9670149;R-HSA-8854044;R-HSA-8854041;R-HSA-8854052;R-HSA-8854051;R-HSA-9942578;R-HSA-5244669;R-HSA-8854071;R-HSA-69299;R-HSA-4086410;R-HSA-5195402;R-HSA-5229194;R-HSA-2993898;R-HSA-2529015;R-HSA-9615937;R-HSA-2529020;R-HSA-2172666;R-HSA-9615945;R-HSA-8961619;R-HSA-8961620;R-HSA-8961632;R-HSA-9943678;R-HSA-9943676;R-HSA-9943675;R-HSA-8961636;R-HSA-9943662;R-HSA-8961651;R-HSA-1214188;R-HSA-9648017;R-HSA-8852354;R-HSA-8961934;R-HSA-8961920;R-HSA-164616;R-HSA-164617;R-HSA-8852362;R-HSA-2294574;R-HSA-2990880;R-HSA-164620;R-HSA-2990882;R-HSA-8961946;R-HSA-2520883;R-HSA-2294580;R-HSA-8853405;R-HSA-2294590;R-HSA-188191;R-HSA-9853878;R-HSA-8961961;R-HSA-2172678;R-HSA-8961952;R-HSA-9686969;R-HSA-8853419;R-HSA-176942;R-HSA-8853429;R-HSA-2429719;R-HSA-8961982;R-HSA-8961972;R-HSA-176956;R-HSA-8853444;R-HSA-5221130;R-HSA-9648089;R-HSA-8961991;R-HSA-9686980;R-HSA-9618378;R-HSA-2980720;R-HSA-113503;R-HSA-9929719;R-HSA-9648114;R-HSA-113504;R-HSA-9929717;R-HSA-162657;R-HSA-9668597;R-HSA-

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Q: What pathway has the most significant “Entities p-value”? Do the most significant pathways listed match your previous KEGG results? What factors could cause differences between the two methods?

Reactome had the most significant “Entities p-value,” and its results mostly matched the KEGG and GO analyses because all three highlighted similar processes like the cell cycle, mitosis, DNA replication, and chromosome segregation. The results are not exactly the same because KEGG, GO, and Reactome use different pathway databases, gene annotations, and analysis methods, and Reactome is usually more detailed than the other two.

Save our results

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write.csv(res, file = "deseq_results2.csv")
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